Using a CSV

Define an arbitrary, curved profile from a width/height trace instead of the rectangular Space opening. A CSV can describe a whole etched profile, not just a simple opening.

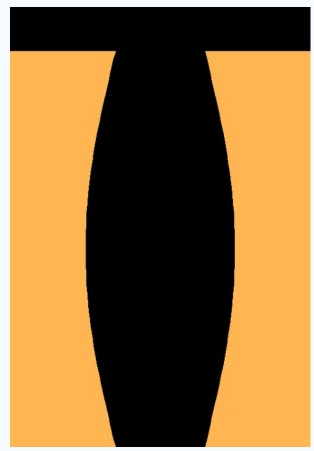
The CSV format

Two columns, one row per point:

- **width** — the full opening width (CD) at that height. A size, always ≥ 0.
- **height** — the vertical position in nm. May be negative (it's a position).
- The profile is drawn **symmetric** about the centerline (±width/2).

width	height
0	0
22	6
36	16
48	40
56	80
62	130
64	180
62	220
58	248
60	262

example_profile.csv — a realistic etched trench



The trace loaded as the profile

Steps

1. Prepare a CSV with columns **width,height** (see example). Save it anywhere.
2. In the **OPENING FROM CSV** section, click **“Load CSV…”** and choose your file.
3. **Placement.** By default the trace is **top-aligned** (its top sits at the stack top, cutting downward). To pin it elsewhere, set **“CSV height=0 at (nm)”** — the row where height = 0 lands that many nm from the stack bottom; negative heights sit below that line.
4. While a CSV is active, **Space** and **Opening depth** are ignored (they gray out); **Top vacuum** still applies.
5. If the curved walls look jagged, raise **Smoothing** (Off / 2× / 4× / 8×) — it de-jags the edges without ever blending colours. Zoom and drag to inspect, use **Measure** to check a CD, then **Save .bmp…** to export.

Tips

- If the material stack is empty, a silicon base is added automatically so the profile is visible.
- Negative width is rejected; ragged rows or blanks are rejected with a clear message.
- Points don't need to be pre-sorted — they're ordered by height internally.